Exercise Sheet 17

The modular convention is $\int_G F(xg)\,dx = \Delta_G(g)^{-1}\int_G F(x)\,dx$ for a left Haar measure dx. Exercise 17.6 is an optional extension.

Exercise 17.1. Let G be the $ax+b$ group of pairs (a,b) with $a>0$, $b\in\mathbb{R}$, written as matrices

$$(a,b)=\begin{pmatrix} a & b\\ 0 & 1\end{pmatrix},\qquad (a,b)(a',b')=(aa',\ b+ab').$$

Its left Haar measure is $dx = a^{-2}\,da\,db$ and $\Delta_G(a,b)=1/a$. Let $A=\{(t,0):t>0\}$ with Haar measure dt/t and $\Delta_A=1$. For $F\in C_c(G)$ put $F^A(xA)=\int_0^\infty F(x(t,0))\,dt/t$.

(a) Show directly that no measure μ on G/A satisfies $\int_G F\,dx=\int_{G/A}F^A\,d\mu$ for all $F\in C_c(G)$.

(b) Show that $\rho(a,b)=a$ satisfies $\rho(xh)=\Delta_A(h)\Delta_G(h)^{-1}\rho(x)$ for $h\in A$.

(c) Identify G/A with $\mathbb{R}$ and compute the action of G. Show that $\int_G F\rho\,dx=\int_{\mathbb{R}}F^A(b)\,db$, so that μ_ρ is Lebesgue measure db.

(d) Compute $\lambda(g,xA)=\rho(gx)/\rho(x)$. For $\varphi\in C_c(\mathbb{R})$, verify the identity

$$\int_{\mathbb{R}}\varphi(g^{-1}\cdot b)\,db=\int_{\mathbb{R}}\varphi(b)\lambda(g,b)\,db.$$

Verify the cocycle identity directly.

Exercise 17.2. Let G, A, $\rho(a,b)=a$ and $\mu_\rho=db$ on $G/A\cong\mathbb{R}$ be as in Exercise 17.1, with $(a,b)^{-1}=(a^{-1},-a^{-1}b)$. Fix $\tau\in\mathbb{R}$ and let $\sigma_\tau(t,0)=t^{i\tau}$, a unitary character of A. The normalised action on $\mathcal{F}_c(\sigma_\tau)$ is $(\Pi(g)f)(x)=(\rho(g^{-1}x)/\rho(x))^{1/2}f(g^{-1}x)$.

(a) Show that every $f\in\mathcal{F}_c(\sigma_\tau)$ has the form $f(a,b)=a^{-i\tau}\varphi(b)$ with $\varphi\in C_c(\mathbb{R})$, and conversely. Show $\|f\|^2=\int_{\mathbb{R}}|\varphi|^2\,db$.

(b) Write $\Pi_\tau(g)$ for the operator $\Pi(g)$ expressed in terms of φ. Show that

$$(\Pi_\tau(a_0,b_0)\varphi)(b)=a_0^{-1/2+i\tau}\,\varphi\Big(\frac{b-b_0}{a_0}\Big).$$

(c) Verify directly that $\Pi_\tau(g_1)\Pi_\tau(g_2)=\Pi_\tau(g_1g_2)$ and that $\Pi_\tau(g)$ is isometric.

(d) Show that $\varphi\mapsto a_0^{i\tau}\varphi((b-b_0)/a_0)$ is not isometric when $a_0\neq 1$.

Exercise 17.3. Let G be a Lie group, H a closed subgroup and (σ,V) a unitary representation of H. A *rho-function* is a continuous $\rho>0$ with $\rho(xh)=\Delta_H(h)\Delta_G(h)^{-1}\rho(x)$. Use the quotient-measure theorem in the following form. For each rho-function ρ there is a unique measure μ_ρ with $\int_G F\rho\,dx=\int_{G/H}F^H\,d\mu_\rho$. Moreover, $F\mapsto F^H$ maps $C_c(G)$ onto $C_c(G/H)$. Let ρ_1,ρ_2 be rho-functions, and let Π_j be the normalised action

$$(\Pi_j(g)f)(x)=\Big(\frac{\rho_j(g^{-1}x)}{\rho_j(x)}\Big)^{1/2}f(g^{-1}x)$$

on $\mathcal{F}_c(\sigma)$ with norm $\|f\|_j^2=\int_{G/H}\|f(x)\|^2\,d\mu_{\rho_j}(xH)$.

(a) Show that $Q = \rho_2/\rho_1$ is right H-invariant and defines a continuous positive function $\bar{Q}$ on G/H.
(b) Show that $\int \varphi \, d\mu_{\rho_2} = \int \varphi \, \bar{Q} \, d\mu_{\rho_1}$ for $\varphi \in C_c(G/H)$.
(c) Let $(Tf)(x) = Q(x)^{-1/2} f(x)$. Show that T maps $\mathcal{F}_c(\sigma)$ onto itself, that $\|Tf\|_2 = \|f\|_1$, and that $T\Pi_1(g) = \Pi_2(g)T$. Conclude that the two induced representations are unitarily equivalent.
(d) For the $ax+b$ group with $H = N$, $\rho_1 = 1$ and $\rho_2(a,b) = a$, identify T.

Exercise 17.4. For $\xi \in \mathbb{R}$ let Π_ξ be the representation of the $ax+b$ group on $L^2((0,\infty), da/a)$ given by

$$(\Pi_\xi(a_0,b_0)\varphi)(a) = e^{i\xi b_0/a}\,\varphi(a/a_0).$$

Continuous functions with $\int_0^\infty |\varphi|^2 \, da/a < \infty$ belong to this space. The action formula applies to them, and their inner products are given by the integral. This follows by compact cutoffs and dominated convergence, using the Radon-measure interpretation in Lecture 16.

(a) Let $\xi = 0$. Show that $\Pi_0(1,c) = I$. Let $(M\varphi)(t) = \varphi(e^t)$. Show that M is isometric from $L^2((0,\infty), da/a)$ to $L^2(\mathbb{R}, dt)$ and that $(M\Pi_0(a_0,b_0)\varphi)(t) = (M\varphi)(t - \log a_0)$.
(b) Let $\varphi(a) = a^{-1}e^{-1/a}$. Compute $\|\varphi\|^2$ and $c_\xi(t) = \langle \varphi, \Pi_\xi(1,t)\varphi\rangle$. Compute $|c_\xi(t)|$.
(c) For $\kappa > 0$ let $(U\varphi)(a) = \varphi(a/\kappa)$. Verify directly that $\langle U\varphi, \Pi_{\kappa\xi}(1,t)U\varphi\rangle = c_\xi(t)$ for the φ of (b).

Exercise 17.5. Let Π_ξ be as in Exercise 17.4. For a strongly continuous unitary representation Π of the $ax+b$ group on a Hilbert space $\mathcal{H}$ and $v \in \mathcal{H}$, put

$$J_\Pi(v) = \int_{\mathbb{R}} t\, e^{-t^2} \, \langle v, \Pi(1,t)v\rangle \, dt.$$

You may use $\int_{\mathbb{R}} e^{-t^2} e^{i\eta t}\, dt = \sqrt{\pi}\, e^{-\eta^2/4}$.

(a) Show that $\int_{\mathbb{R}} t\, e^{-t^2} e^{i\eta t}\, dt = i\,\frac{\sqrt{\pi}}{2}\,\eta\, e^{-\eta^2/4}$.
(b) Show that $|J_\Pi(v) - J_\Pi(v')| \le (\|v\| + \|v'\|)\,\|v - v'\|$.
(c) For $\varphi \in C_c((0,\infty))$ show that

$$J_{\Pi_\xi}(\varphi) = i\,\frac{\sqrt{\pi}}{2}\int_0^\infty \frac{\xi}{a}\, e^{-\xi^2/(4a^2)}\, |\varphi(a)|^2\, \frac{da}{a}.$$

(d) Let Σ and Σ' be strongly continuous unitary representations of the $ax+b$ group on Hilbert spaces $\mathcal{H}$ and $\mathcal{H}'$. Show that $J_{\Sigma'}(Uv) = J_\Sigma(v)$ for $v \in \mathcal{H}$ if $U\colon \mathcal{H} \to \mathcal{H}'$ is a unitary intertwiner from Σ to Σ'.
(e) Deduce that Π_1, Π_{-1} and Π_0 are pairwise inequivalent.

Exercise 17.6 (Optional extension). Let π be the case $\tau = 0$ of Exercise 17.2. It acts on $L^2(\mathbb{R}, db)$ by

$$(\pi(a_0, b_0)\varphi)(b) = a_0^{-1/2}\,\varphi\Big(\frac{b - b_0}{a_0}\Big).$$

Let $\Pi_{\pm 1}$ be as in Exercise 17.4. Write $\mathbb{R}_+ = (0, \infty)$ and $\mathbb{R}_- = (-\infty, 0)$, both with Lebesgue measure $d\eta$. Use without proof the following facts. The Fourier transform

$$\hat{\varphi}(\eta) = (2\pi)^{-1/2} \int_{\mathbb{R}} \varphi(b) e^{i\eta b}\, db$$

on $C_c(\mathbb{R})$ extends to a unitary operator $\mathcal{F}$ of $L^2(\mathbb{R})$. Also, $C_c(\mathbb{R} \setminus \{0\})$ is dense in $L^2(\mathbb{R})$. We use the usual integral inner product on continuous square-integrable functions, obtained by compact cutoffs. For $\varphi \in C_c(\mathbb{R})$, the vector $\mathcal{F}\varphi$ is represented by the continuous function $\hat{\varphi}$ above. The action formula in (b) extends to such functions by density and dominated convergence.

(a) Show that $\widehat{\pi(a_0, b_0)\varphi}(\eta) = a_0^{1/2} e^{i\eta b_0} \hat{\varphi}(a_0\eta)$.

(b) Show that $L^2(\mathbb{R}) = L^2(\mathbb{R}_+) \oplus L^2(\mathbb{R}_-)$. Show that both summands are invariant under the operators

$$(\hat{\pi}(a_0, b_0)u)(\eta) = a_0^{1/2} e^{i\eta b_0} u(a_0\eta).$$

(c) For a function u on $\mathbb{R}_\pm$ put

$$(V_\pm u)(a) = a^{-1/2} u(\pm 1/a) \qquad (a > 0).$$

Show that V_+ is a unitary map from $L^2(\mathbb{R}_+)$ onto $L^2((0, \infty), da/a)$. Show the same for V_- on $L^2(\mathbb{R}_-)$. Show that $V_\pm \hat{\pi}(g) = \Pi_{\pm 1}(g) V_\pm$. Conclude that $\pi \cong \Pi_1 \oplus \Pi_{-1}$.